

MORE ON ENCRYPTION MODES AND THEIR SECURITY

Analysis of the CTRC

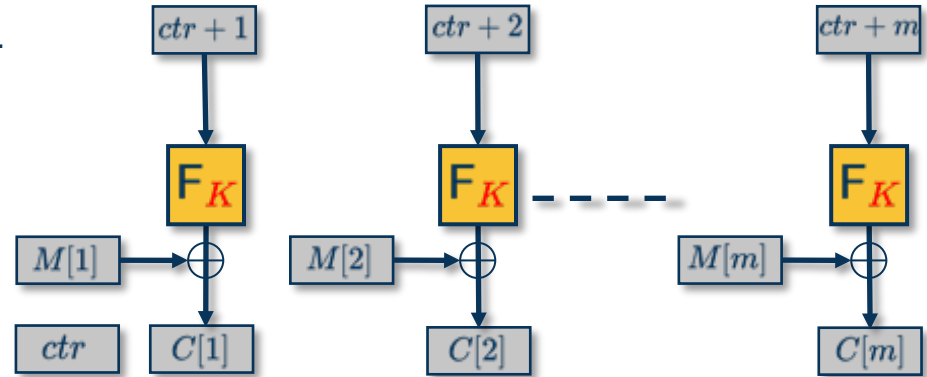
Let $F : \{0, 1\}^k \times \{0, 1\}^\ell \rightarrow \{0, 1\}^\ell$ be a function family.

CTRC = $(\{0, 1\}^k, \mathcal{E}, \mathcal{D})$:

A current counter is ctr maintained as a state

Encryption algorithm $\mathcal{E}$

ctr is initially 0^ℓ



The scheme is used to encrypt at most 2^ℓ blocks (so that the counter does not wrap around)

- How good is the scheme?
- The flaws seem hard to find.
- Q. But may be they exist and we just don't see them?
- A. The mode is as good as it can be and we can prove it.

How to Prove Security

To prove that scheme S is x -secure assuming scheme T is y -secure, use contraposition:

- scheme T is y -secure $\Rightarrow$ scheme S is x -secure is equivalent to
- scheme S is not x -secure $\Rightarrow$ scheme T is not y -secure is equivalent to
- there exists an efficient x -adversary $\mathcal{A}$ breaking $S \Rightarrow$ there exists an efficient y -adversary $\mathcal{B}$ breaking T
 1. We assume $\mathcal{A}$ exists, and then construct $\mathcal{B}$, who can use $\mathcal{A}$.
 2. We show that $\mathcal{B}$'s y -advantage is not too small when $\mathcal{A}$'s x -advantage is not too small.
 3. We show that $\mathcal{B}$'s resources are reasonable when $\mathcal{A}$'s resources are reasonable.

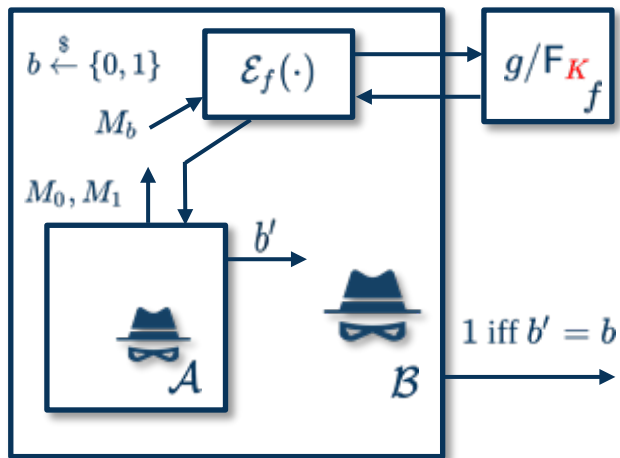
Security of CTRC

- ◆ **Theorem:** For any efficient adversary $\mathcal{A}$ there exists an adversary $\mathcal{B}$ with similar resources such that

$$\text{Adv}_{\text{CTRC}}^{\text{ind-cpa}}(\mathcal{A}) \leq 2 \cdot \text{Adv}_F^{\text{prf}}(\mathcal{B})$$

- ◆ **Proof idea:**
 - One can present an adversary $\mathcal{B}$ who needs to distinguish whether it is given the oracle access to a truly random function or an instance of F .
 - $\mathcal{B}$ will use $\mathcal{A}$'s ability to break the CTRC encryption scheme. $\mathcal{B}$ will run $\mathcal{A}$ as a subroutine, simulating the cpa-cg experiments for it. $\mathcal{B}$ will answer $\mathcal{A}$'s oracle queries using its own oracle. Finally, if $\mathcal{A}$ wins, $\mathcal{B}$ will win.
 - The crucial point is that when inside of $\mathcal{B}$'s oracle there is a random function, $\mathcal{A}$ had no advantage, as it's basically given the OTP.

- Proof:** Let $\mathcal{A}$ be any cpa-cg adversary attacking CTRC. We present a prf adversary $\mathcal{B}$:



g is a random instance of $\text{Func}(\ell, L)$

$\mathcal{B}$ can simulate the CTRC encryption algorithm because it makes only “oracle” use of the underlying function F

Adversary $\mathcal{B}^f$

$b \xleftarrow{\$} \{0, 1\}$

Run adversary $\mathcal{A}$, replying to its oracle queries as follows,

When $\mathcal{A}$ makes an oracle query (M_0, M_1) do

$C \xleftarrow{\$} \mathcal{E}_f(M_b)$

Return C to $\mathcal{A}$ as the answer,

Until $\mathcal{A}$ stops and outputs a bit b'

If $b' = b$ then return 1 else return 0

- Let us analyze $\mathcal{B}$. One can show that

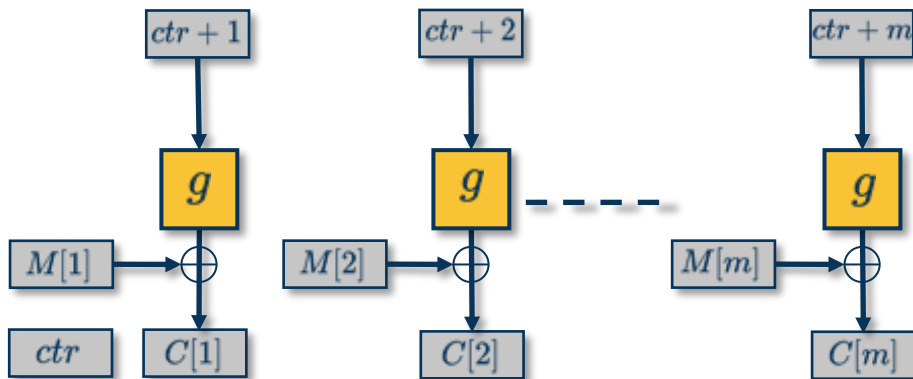
$$\begin{aligned}\text{Adv}_F^{\text{prf}}(\mathcal{B}) &= \Pr[\mathcal{B} \Rightarrow 1 \text{ in } \mathbf{Exp}_F^{\text{prf-1}}] - \Pr[\mathcal{B} \Rightarrow 1 \text{ in } \mathbf{Exp}_F^{\text{prf-0}}] \\ &= \Pr[\mathbf{Exp}_{\text{CTRC}[F]}^{\text{cpa-cg}} \Rightarrow 1] - \Pr[\mathbf{Exp}_{\text{CTRC}[\text{Func}(\ell, L)]}^{\text{cpa-cg}} \Rightarrow 1] \\ &= \frac{1}{2} \cdot \text{Adv}_{\text{CTRC}[F]}^{\text{ind-cpa}}(\mathcal{A}) + \frac{1}{2} - \frac{1}{2} \cdot \text{Adv}_{\text{CTRC}[\text{Func}(\ell, L)]}^{\text{ind-cpa}}(\mathcal{A}) - \frac{1}{2}\end{aligned}$$

- $\text{CTRC}[F]$ is the standard CTRC mode
- $\text{CTRC}[\text{Func}(\ell, L)]$ is the CTRC mode a random function is used.

- We will show that $\text{Adv}_{\text{CTRC}[F, \ell, L]}^{\text{ind-cpa}}(\mathcal{A}) = 0$ and the statement of the theorem will follow after we justify the resources of $\mathcal{B}$:

$\mathcal{B}$ runs $\mathcal{A}$ and compares two bits. Its number of queries is equal or less than the number of blocks in all $\mathcal{A}$'s queries. The length of $\mathcal{B}$'s queries is the number of queries times ℓ (the length of the block). Clearly, if $\mathcal{A}$ has reasonable resources, then $\mathcal{B}$ does too.

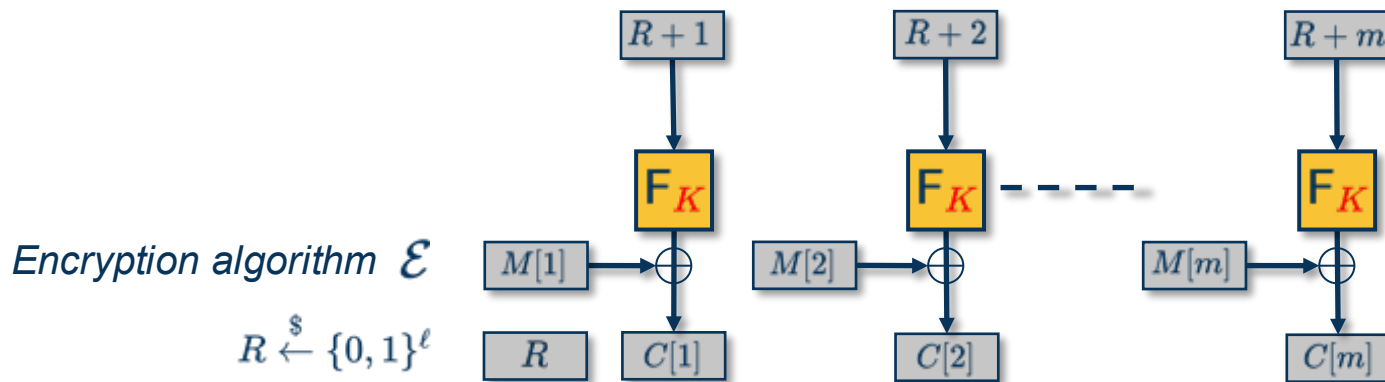
- $\text{Adv}_{\text{CTRC}[\text{Func}[\ell, L]]}^{\text{ind-cpa}}(\mathcal{A}) = 0$ because all the values corresponding to the red dots on the picture below are random and independent (since they are the results of a random function applied to distinct points) and thus $C[1], \dots, C[m]$ are also random values, independent from the adversary $\mathcal{A}$'s challenge bit.



Analysis of the CTR\$

Let $F : \{0, 1\}^k \times \{0, 1\}^\ell \rightarrow \{0, 1\}^L$ be a function family.

$\text{CTR\$} = (\{0, 1\}^k, \mathcal{E}, \mathcal{D}) :$



Security of CTR\$

- ◆ **Theorem:** For any adversary $\mathcal{A}$ there exists an adversary $\mathcal{B}$ with similar resources such that

$$\text{Adv}_{\text{CTR\$}}^{\text{ind-cpa}}(\mathcal{A}) \leq 2 \cdot \text{Adv}_{\text{F}}^{\text{prf}}(\mathcal{B}) + \frac{\mu_{\mathcal{A}}^2}{\ell^2 \cdot 2^\ell}.$$

where μ is the total number of bits $\mathcal{A}$ sends to the oracle.

- ◆ What does the security statement tell us?

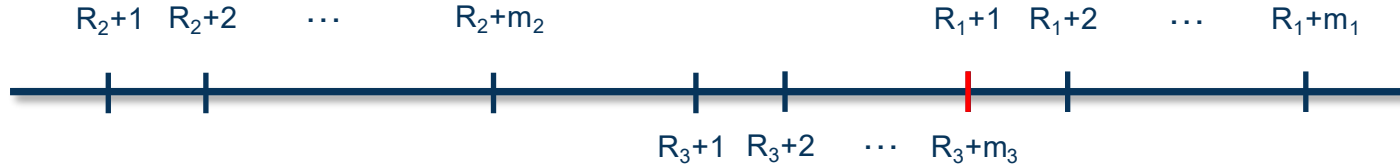
- Let F be AES, $\ell = L = 128$. Assume one encrypts $q = 2^{30}$ messages, 1 Kb each (2^{13} bits),

recall $\text{Adv}_{\text{F}}^{\text{prf}}(\mathcal{B}) \leq \approx \frac{q_{\text{AES}}^2}{2^{128}}$

$$\begin{aligned} \text{Adv}_{\text{CTR\$}}^{\text{ind-cpa}}(\mathcal{A}) &\leq \approx 2 \cdot \frac{q_{\text{AES}}^2}{2^{128}} + \frac{\mu_{\mathcal{A}}^2}{\ell^2 \cdot 2^\ell} = \frac{3 \cdot \mu^2}{\ell^2 \cdot 2^{128}} \\ &\leq \frac{4 \cdot 2^{43 \cdot 2}}{128^2 \cdot 2^{128}} = \frac{1}{2^{54}} \end{aligned}$$

- ◆ **Proof idea:** As in the proof of the previous theorem.

Intuition for the Additive Term

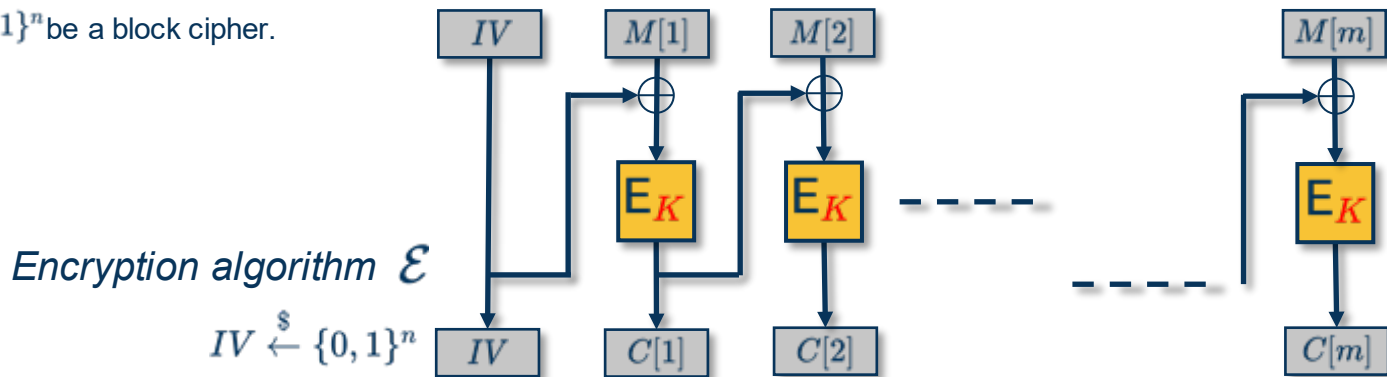


- R_i - IV of query i
- m_i – number of blocks in each of the messages of query i
- If there are no collisions then security is as good as of CTRC
- The probability of a collision relates to the birthday paradox
- It is very small for practical parameters

Security of CBC\$

Let $E : \{0, 1\}^k \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher.

$\text{CBC\$} = (\{0, 1\}^k, \mathcal{E}, \mathcal{D}) :$



- Theorem:** For any adversary $\mathcal{A}$ there exists an adversary $\mathcal{B}$ with similar resources such that

$$\text{Adv}_{\text{CBC\$}}^{\text{ind-cpa}}(\mathcal{A}) \leq 2 \cdot \text{Adv}_{\text{E}}^{\text{prf}}(\mathcal{B}) + \frac{\mu_{\mathcal{A}}^2}{n^2 \cdot 2^n}$$

Summary of Results so far

- ◆ AES is believed to be a PRF
- ◆ ECB (and any deterministic stateless encryption scheme) is not IND-CPA
- ◆ CBC is not IND-CPA
- ◆ CBC\$, CTRC, CTR\$ are IND-CPA when used with AES
- ◆ The security guarantees for CTRC are slightly better than for CBC\$, CTR\$

Did We Get All We Wanted?

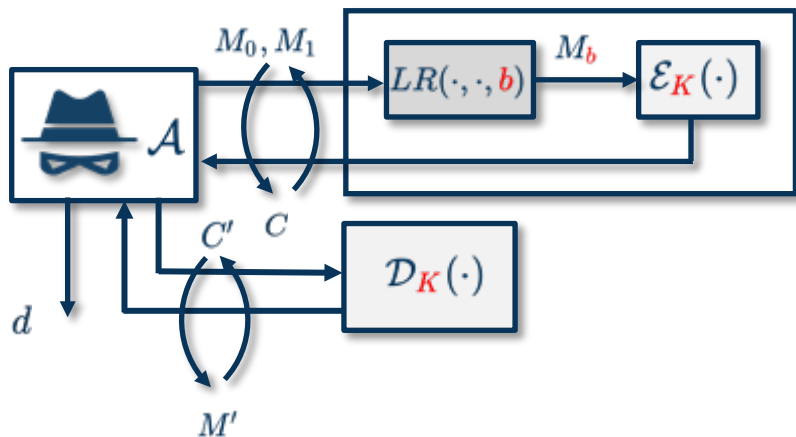
- ◆ Is IND-CPA security definition strong enough (does it take into account all the bad things that can happen?)
- ◆ An adversary wants to win: to get some partial information about the plaintext from a challenge ciphertext
- ◆ What if the adversary can make the receiver to decrypt other ciphertexts of the adversary's choice, learn the plaintexts and this helps it to win?
- ◆ Our definition didn't consider such "chosen-ciphertext" attacks

Indistinguishability under chosen-ciphertext attacks

Fix $\mathcal{SE} = (\text{KeySp}, \mathcal{E}, \mathcal{D}), K \xleftarrow{\$} \text{KeySp}$

For an adversary $\mathcal{A}$ and a bit b (either 0 or 1) consider an experiment $\text{ind-cca-}b$, i.e., consider two experiments, ind-cca-0 (“left”) and ind-cca-1 (“right”).

In each experiment, $\mathcal{A}$ is given access to the “left-right” encryption oracle that takes two messages and returns encryption of M_b under K , and decryption oracle. $\mathcal{A}$ cannot query the decryption oracle on ciphertexts from the “left-right” encryption oracle.



The IND-CCA advantage of $\mathcal{A}$ (denoted as $\text{Adv}^{\text{ind-cca}}(\mathcal{A})$) is $\Pr[\mathcal{A} \Rightarrow 0 \text{ in ind-cca-0 exp.}] - \Pr[\mathcal{A} \Rightarrow 0 \text{ in ind-cca-1 exp.}]$

A symmetric encryption scheme $\mathcal{SE}$ is indistinguishable under chosen-ciphertext attacks (IND-CCA secure) if for any adversary $\mathcal{A}$ with “reasonable” resources its ind-cca advantage is “small” (close to 0).

- ◆ IND-CCA implies IND-CPA
 - I.e., IND-CCA is the stronger notion
 - I.e., any scheme that is IND-CCA is also IND-CPA

Are the modes we studied IND-CCA?

- ◆ ECB and CBC_{CC} are not IND-CCA. Why?
- ◆ CBC_{\$}, CTR_C, CTR_{\$} are not IND-CCA either.
 - Proof.

What to do?

- ◆ We will find a way to construct IND-CCA schemes
- ◆ But we have to study more tools first